

Chapter 107 — Urodynamic and Videourodynamic Evaluation of the Lower Urinary Tract

- **Lower urinary tract (LUT) dysfunction** produces lower urinary tract symptoms (LUTS) that impair quality of life, but it can also be "**silent**" — asymptomatic yet placing the upper tracts at risk.
- **Urodynamics (UDS)** is the dynamic study of the transport, storage and evacuation of urine: a set of tests, used singly or together, to **diagnose suspected dysfunction and assess the effect of known dysfunction**.
- The LUT has **two essential functions: low-pressure storage** (protects the kidneys, ensures continence) and **voluntary evacuation** (socially acceptable voiding without leakage or overdistension).
- Disruption produces either **bothersome LUTS** (frequency, urgency, urgency urinary incontinence (UUI), slow or interrupted stream) or **harmful sequelae** — urinary tract infection (UTI) and renal failure.
- International Continence Society (ICS) usage distinguishes **UDS studies** (the tests performed) from **UDS observations** (what is seen during testing).
- **What has changed:** not the principles or technique, but the **abandonment of absolute reliance on UDS**. Studies are now tailored to specific populations within a disease state, and used when available data are insufficient to choose treatment — to inform, reassure or uncover other pathology.
- The American Urological Association (AUA) now favours **disease-specific guidelines** — overactive bladder (OAB), neurogenic lower urinary tract dysfunction (NLUTD), stress urinary incontinence (SUI) — over the older **UDS-specific guideline** designed to help select tests after symptom characterisation.
- **Children:** principles and equipment are identical, but in children UDS **establishes a baseline** (anatomical and neurological abnormalities make function variable and unpredictable) so that change with treatment and growth can be tracked, and informs caregivers and patient about current and future function. Testing must still be **relevant, reliable and reproducible**. Adult conditions only are covered here.

ROLE OF URODYNAMICS IN CLINICAL PRACTICE

- **Level 1 evidence for specific "indications" is limited**, for structural reasons:
 - randomised controlled trials (RCTs) are hard to justify where lower-level evidence strongly supports utility and **empirical treatment is potentially harmful or life-threatening** (e.g. NLUTD);
 - symptoms have many causes, so **homogeneous populations are hard to study**;
 - high-level data on UDS in counselling and shared decision-making are absent;
 - cost-effectiveness questions (e.g. who benefits from early surgery for bladder outlet obstruction (BOO) from benign prostatic hyperplasia (BPH)) need complex modelling or expensive long-term studies.
- UDS is **most useful when history, examination and simple tests are insufficient** — two general scenarios:
 - **Diagnosis** of the condition causing symptoms (LUTS, incontinence).
 - **Determining the impact of a disease capable of serious, irreversible upper- and lower-tract damage** — spinal cord injury (SCI), multiple sclerosis (MS), radiation cystitis. **Profound abnormalities may exist with few symptoms**.
- Rather than a list of indications, think of the **broader clinical roles** of UDS:
 - demonstrate the **pathophysiology** behind presenting signs or symptoms;
 - identify **contributing factors** and their relative importance;
 - obtain information on **other aspects of LUT function**, symptomatic or not;
 - **predict upper-tract consequences** (Box 107.1);

- **predict treatment outcome and side effects**, aiding pre-operative counselling;
- **confirm the effect or mode of action** of an intervention, especially a new one;
- **explain failure** of previous treatment.
- **Governing principles of interpretation:**
 - the goal of a symptom-driven study is to **reproduce the symptom and identify its cause**;
 - **customise** the study — decide the questions first, then design the study to answer them;
 - UDS is an **unnatural setting**: an abnormality recorded without reproduction of symptoms is **not necessarily diagnostic**;
 - **failure to record an abnormality does not exclude it** (e.g. no detrusor overactivity (DO) seen in a patient with UUI);
 - **not every observation is clinically significant** — interpret with history, comorbidity, **postvoid residual (PVR)** and **frequency-volume charts**.
- **Technical care in data collection remains the cornerstone** of a good study, despite advances in recording and processing.

BOX 107.1 URODYNAMIC RISK FACTORS

Findings that are potentially dangerous and **usually require intervention** to prevent upper- and lower-tract decompensation:

1. **Impaired compliance**
2. **Detrusor-external sphincter dyssynergia (DESD)**
3. **Detrusor-internal sphincter dyssynergia (DISD)**
4. **High-pressure DO present throughout filling**
5. **Elevated detrusor leak point pressure (> 40 cm H₂O)**
6. **Poor emptying with high storage pressures**

- **What the box is for:** every item is a route to **sustained high intravesical pressure**. The trap is to treat these as symptom findings — they mandate treatment **even when the patient is asymptomatic**.

KEY POINTS

- UDS is the **dynamic study of the transport, storage and evacuation of urine**, made up of tests used singly or together.
- Guidelines assist **selection of tests after evaluation and symptom characterisation**.

Concept — *UDS earns its place in two situations only: when the diagnosis cannot be made without it, and when the disease can silently destroy the upper tracts. Every clinical role follows from answering a pre-defined question, never from the mere presence of symptoms.*

CONDUCTING A URODYNAMIC STUDY: PATIENT AND TECHNICAL FACTORS

Functional Classification of Voiding Dysfunction

- **The single most important concept** for setting up, conducting and interpreting UDS — the **Wein functional classification (1981)**.
- **Functional division:**
 - **storage dysfunction** — failure to store;
 - **emptying dysfunction** — failure to empty;

- **combined** — failure of both.

- **Anatomical subclassification** — the examinable pairing of site against direction:

Site	Overactive	Underactive
Bladder	Failure to store	Failure to empty
Bladder outlet	Failure to empty	Failure to store

- Plus **combined bladder and outlet dysfunction**.
- **Why it matters:** it generates the questions before the study, anticipates findings, and **organises treatment options**. It prevents the clinician becoming lost in neurophysiology instead of thinking practically: *is the problem bladder, outlet or both — and storage, emptying or both?*

Concept — *The classification is a **two-by-two of site against direction**: overactivity of the bladder and underactivity of the outlet both cause storage failure, and the reverse pair causes emptying failure. Each study is designed to locate the patient in that grid.*

Preparing for Urodynamic Studies: Clinician, Patient, and Facility

Defining the Clinical Question

- **Three rules before any invasive study:** **decide the questions** → **design the study** to answer them → **customise** as necessary.
- **Symptoms alone are not an indication** — the information required must be defined in advance, informed by expert opinion, history, medication and examination.
- **Medication status must be decided and documented in advance** (antimuscarinics, β_3 -adrenergic agonists):
 - **to assess a drug's therapeutic effect** → test **on** a regular dosing schedule;
 - **to uncover the aetiology** of symptoms treated empirically → consider **stopping** the drug first (highest diagnostic yield).
- **Pre-invasive work-up** recommended by the ICS: **validated symptom and bother score**, **3-day frequency-volume chart**, and **non-invasive uroflowmetry and PVR**. The International Urogynecological Association (IUGA) proforma begins with an exhaustive pre-study checklist.
- **Why:** a normal free flow and low PVR beforehand allow an abnormal flow during invasive testing to be recognised as **probably artefactual**.

Patient Preparation

- Explain **why the test is done**, **how it may change treatment** and **what to expect**.
- UDS is **generally well tolerated** — anxiety, discomfort and embarrassment are minimal to moderate, and most patients find it the same as or better than expected; most would repeat it. **Commonest complaint afterwards: pain on micturition**.
- **Younger patients report more pain and anxiety** → need more reassurance.
- **Simple environmental measures** — dimmed lighting and music (trend only), and **lavender aromatherapy** (reduced anxiety in an RCT) — may help.
- **Explanatory pamphlets have been disappointing**; well-trained, attentive, supportive staff matter most.
- **Urinalysis before every invasive study** (level 4 evidence).
- **Symptomatic UTI** → **postpone** until treatment is complete.
- **"Index" patients** (no UTI risk factors): the AUA Best Practice Policy (BPP) Statement on antimicrobial prophylaxis — **routine prophylaxis not indicated**.

- **Non-index patients:** the Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (SUFU) BPP panel recommends a **single dose within 1 hour of UDS:**
 - **first choice:** oral trimethoprim–sulfamethoxazole;
 - **alternatives:** first- or second-generation cephalosporin, amoxicillin–clavulanate, intravenous aminoglycoside ± ampicillin, fluoroquinolones (level 3).

TABLE 107.1 RISK FACTORS AND RECOMMENDATIONS FOR ANTIBIOTICS BEFORE INVASIVE URODYNAMIC STUDIES

Risk factor	Prophylaxis?	Level
Pre-procedure testing — urinalysis in all; positive urinalysis without UTI → prophylaxis	Urinalysis all	4
Current symptomatic UTI	Cancel until treated	4
Presumed normal genitourinary anatomy, no risk factors (risk factor discovered during study → give immediately after)	No	1
NLUTD	Yes	4
BOO and/or clinically important elevated PVR (any cause)	Yes	4
Age > 70 years	Yes	2
Asymptomatic bacteriuria	Yes	4
Immunosuppression — drugs, chronic steroids, innate; especially renal transplant	Yes	4
Chronic catheters — urethral or suprapubic indwelling, condom catheter, clean intermittent catheterisation (CIC)	Yes	4
Orthopaedic joint implant at increased risk of joint infection or bacteraemia	Yes	3
Diabetes mellitus	No	4
Recent prolonged hospitalisation (no other risk factor)	No	4
Diet and nutrition, including obesity	No	4
Menopausal status	No	2
History of recurrent UTI	No	4
Gender	No	4
Prior urological surgery	No	4
Cardiac valvular disease	Not routinely	3

From Cameron AP, Campeau L, Brucker BM, et al.: Best practice policy statement on urodynamic antibiotic prophylaxis in the non-index patient, *Neurourol Urodyn* 36:915–926, 2017.

- **What the table is for:** separating factors that genuinely raise the risk of **post-instrumentation infection or bacteraemia with consequences** (stasis, bacteriuria, foreign material, immune compromise, prostheses, age > 70) from factors that merely sound high-risk.
- **Pattern:** "yes" factors involve **stasis, colonisation, catheters, immune compromise or implants**; "no" factors are general medical states.
- **Level of evidence:** almost all level 4; only the **normal-anatomy "no"** is level 1, and **age > 70** and **menopausal status** level 2.
- **Traps:** recurrent UTI, diabetes and cardiac valve disease are "no"; asymptomatic bacteriuria is "yes" (unlike most urological contexts, where it is left untreated).

Facility Setup

- **Dedicated, quiet, private room** without interruptions — recreating a natural environment is already difficult.
- **Size:** enough to lie, stand and sit on a commode; room for **wheelchairs, walkers, scooters and hoists** (many patients have neurological mobility limits). **Videourodynamics (VUDS) needs more space** for fluoroscopy, and local radiation regulations apply.
- **Equipment ready beforehand:** **pessary kits, Scopette swabs and speculums** for pelvic organ prolapse (POP) reduction; a **step stool** (male urethral voiding images, or standing female stress views when the C-arm has limited minimum height); **large catheters** to drain a bladder when pre-invasive PVR is high.

Additional Factors

- An unanswered question → **repeat in the same session or later**; a **second filling and/or voiding cycle** is widely advocated.
- Even with best practice UDS **does not answer every question** — defining the information needed beforehand keeps unanswered questions to a minimum.
- **Proper performance with careful attention to technical detail** is the prerequisite for accurate interpretation (ICS Good Urodynamic Practices).
- **ICS terminology** should be used by all performing and interpreting UDS, for consistent reporting. **"Urodynamic observations"** = observations made and measured during the test itself. A list of common terms is given in **Box 107.2**; the ICS maintains an online glossary drawn from **19 publications**, including condition-specific terminology for **female pelvic floor dysfunction** and **adult neurogenic** patients.

Concept — *Preparation exists so that the invasive study answers a question decided in advance: the medication status, the pre-invasive flow and PVR, the antibiotic decision and the room itself are all chosen to make the result interpretable.*

COMPONENTS OF THE URODYNAMIC STUDY

- Components can be used **individually or combined**; all are parts of multichannel UDS or VUDS.

Postvoid Residual.

- **Excellent assessment of emptying**; by **ultrasound (bladder scan) or catheterisation** — method must be specified.
- **Elevated PVR shows that emptying is impaired, not why.**
- **Confirm a raised PVR with a second measurement**, especially if the patient had no true desire to void.

Uroflowmetry.

- **Rate of urine flow over time**; also an assessment of emptying. Reportable parameters:
 - **voided volume (VV)**, mL; **flow rate (Q)**, mL/s;
 - **maximum flow rate (Q_{max})** and **average flow rate (Q_{ave})**, mL/s;
 - **voiding time** (total micturition time) vs **flow time** (time during which flow occurred);
 - **time to maximum flow** (onset to Q_{max});
 - **voiding efficiency** = $VV / (VV + PVR) \times 100$.
- **Curve shape matters: normal = bell-shaped** (Fig. 107.1). Interpretation is partly subjective.
- **Pattern → suggestion, not diagnosis:**
 - **interrupted or straining** → detrusor underactivity (or obstruction, or abdominal voiding);
 - **flattened (plateau)** → **fixed obstruction**.

- **Without detrusor pressure, the cause cannot be identified.**
- Standalone test: **preferred position** (sitting or standing); ask whether the flow was **representative**; compare VV with the frequency-volume chart; **record voiding position**.

Cystometry (Filling Cystometrograph).

- **Pressure-volume relationship of the bladder during filling.** Filling runs from the start of infusion to "permission to void", which marks **maximum cystometric capacity (MCC)**.
- A single bladder line records **vesical pressure (Pves)** only, and cannot distinguish a detrusor rise from an **abdominal pressure (Pabd)** rise.
- **Recommended:** measure **both Pves and Pabd** (rectal or vaginal catheter) and derive the **detrusor pressure (Pdet)**:
 - $Pdet = Pves - Pabd$ (Fig. 107.2).

Pelvic Muscle Electromyography.

- **Electromyography (EMG)** records potentials from depolarising muscle membranes — here the **striated perineal sphincter muscles** — during **filling and emptying**.
- **ICS standard:** **two skin-surface perineal electrodes with an appropriate reference**. Alternatives: vaginal probe, anal probe, perineal needle.

Urethral Pressure Profilometry.

- **Urethral pressure profilometry (UPP):** intraluminal pressure along the urethra, recorded by **withdrawing a pressure sensor**.
- **Urethral pressure = fluid pressure needed just to open a closed urethra.**

Pressure-Flow Studies.

- **Pressure-flow studies (PFS):** relationship between **Pdet and flow rate during voiding**, flow measured simultaneously by uroflowmeter.
- **Voiding phase:** from **permission to void** (or onset of uncontrollable voiding) until the **patient considers voiding finished**.

Figure Legends

- **Fig. 107.1 — Uroflow curves:** (A) **normal bell-shaped**; (B) **flattened — usually obstruction**; (C) **interrupted/straining** — impaired contractility, obstruction or abdominal straining. (Wein, English and Whitmore, 1988; Boone and Kim, 1998.)
- **Fig. 107.2 — Single-channel vs multichannel CMG:** (A) Pves alone shows multiple spikes whose origin cannot be determined; (B) adding Pabd shows the spikes were **abdominal (movement, coughing)** — the **subtracted Pdet is flat**. **Standard adult fill rates: 30–100 mL/min.** (Nitti, 1998.)

Concept — *Non-invasive tests (PVR, uroflow) tell you **that** emptying is abnormal; only subtracted detrusor pressure tells you **why**. The whole architecture of multichannel UDS exists to separate the detrusor's contribution from the abdomen's.*

URODYNAMIC EQUIPMENT

Urodynamic Systems

- Current systems are **computer-based digital** platforms — storage, post-processing, upgrades, electronic record integration.
- Choice depends on **patient population and disease spectrum, space, convenience, data handling**.

- **Minimum recommended: multichannel — Pves, Pabd (and subtracted Pdet) and flow;** optional EMG and urethral pressure channels.
- **The clinician remains the most important constant** in data collection and interpretation.
- **Infusate:** a liquid approximating urine — **normal saline or radiographic contrast. Gas (carbon dioxide) is not recommended.**
- **VUDS** adds fluoroscopy: costly, time-consuming, needs a larger room — ideally in **centres with an interest in complex dysfunction;** it may be the **test of choice in advanced NLUTD.**

Signal Transmission and Transducers

COMPARISON: PRESSURE TRANSDUCER SYSTEMS

Feature	External water-filled	Air-charged catheter	Microtip / fibreoptic
Principle	Fluid column from catheter to external strain-gauge transducer	Air-filled balloon round catheter → air lumen → external transducer; circumferential reading	Transducer within the catheter tip , cable to machine
Reference level	Must sit at symphysis pubis; re-level when position changes	None needed — no hydrostatic effect	—
Flushing / air	No air bubbles — lucent tubing so air is visible	No flushing	—
Motion artefact	Significant motion and hydrostatic artefact	Much less	—
Damping	Underdamped — resonates ~10 Hz, attenuates > 19 Hz	Overdamped — attenuates > 3 Hz	—
Cost / reuse	—	Disposable, single-use	Expensive, reusable — sterilise each use
Standards	Most nomograms derived with this system	Reference values not established	—

- **Frequency trade-off:** most urodynamic signals are < 3 Hz, so air-charged damping removes noise — but **rapid events (coughs) contain higher frequencies** and may be blunted. The system must be matched to the application.
- **Water-filled and air-charged systems are not interchangeable:** simultaneous recordings differ, with **Pdet varying by up to 10 cm H₂O. Pattern analysis is unaffected**, but **absolute values** matter in PFS and UPP.
- Air-charged vs microtip for urethral pressure and **Valsalva leak point pressure (VLPP):** one study found comparable performance, another **systematically higher air-charged readings.**
- **7th International Consultation on Incontinence (ICI):** air-charged catheters are an **acceptable alternative for female urethral closure pressure**, but **no reference values; not widely tested in men, children or NLUTD;** different catheters record different values — **users must verify equivalence to their own system.**

Concept — Every transducer is a compromise between **artefact and fidelity**: the fluid column that makes water-filled systems the reference standard also makes them sensitive to height and motion, and the damping that makes air-charged systems clean also blunts fast signals. Absolute numbers transfer between systems badly; patterns transfer well.

Uroflowmeters

COMPARISON: UROFLOWMETER TYPES

Type	Principle	Advantage	Limitation
Weight transducer (load cell) — commonest	Voided weight, differentiated with respect to time	Simple, reliable, accurate regardless of where the stream strikes	Urine density must be set
Rotating disc — commonest	Power needed to keep the disc at constant speed $\propto$ flow	Reliable, accurate; direct flow measurement, no differentiation	—
Electronic dipstick (capacitance)	Capacitance $\propto$ accumulated volume $\rightarrow$ differentiated	—	—

- **Electronic flowmeters are precise enough for clinical use**, with small errors in VV and flow.
- **Infusate specific gravity alters calculated flow** — most systems calibrate for contrast.

Electromyography

- The electrode must lie **in or near the muscle**: surface, needle, anal or vaginal plug, or urethral catheter-mounted.
- **Surface electrodes** are self-adhesive patches over the **anal sphincter**; **external anal sphincter activity mirrors external urethral sphincter EMG except in some neurological diseases**.

Surface electrode	Concentric needle electrode
Comfortable, convenient	Uncomfortable , worse with repeated placement
Inferior signal ; must be precisely placed	Superior signal — preferred by most for striated sphincter
Stable	Easily dislodged ; patients tolerate replacement poorly

- **Electrode choice depends on the question**; interpret EMG in the **overall clinical context**.

KEY POINTS

- **Define the reason first**; the result must be capable of guiding treatment.
- UDS locates dysfunction as **bladder vs outlet, storage vs emptying**, by **reproducing symptoms**.
- Interpret with **history, PVR and frequency-volume charts**; use **ICS practice and terminology**.
- **Trained, supportive staff** and a fully prepared patient are essential.

THE URODYNAMIC STUDY: ANALYSIS AND INTERPRETATION

- Separate the study into **filling/storage** and **voiding** phases — mirrors the functional classification.
 - **Filling/storage**: CMG, **provocative testing** (abdominal leak point pressure (ALPP)), UPP.
 - **Voiding**: **contractility, outlet resistance and sphincter coordination** — by PFS, EMG and, in some cases, fluoroscopy.

Filling and Storage Phase

Normal Filling and Storage

- **Normal bladder stores at low pressure and does not contract involuntarily**; at capacity or when voiding is desired, pressure rises with a **voluntary detrusor contraction** — **preceded by external sphincter relaxation**.

- **Pdet stays near zero throughout filling** — constant low baseline, **no involuntary contractions** (Fig. 107.3). ICS term: **normal detrusor function** (formerly "stable").
- **Pabd by rectal, vaginal or stomal catheter** — data on vaginal and stomal lines are limited, so **use a rectal catheter where possible**.
- **Subtraction distinguishes a true detrusor rise (contraction or loss of compliance) from abdominal straining** — critical when detrusor rises are small or coincide with abdominal changes.

Abnormalities of Bladder Filling: Detrusor Overactivity and Impaired Compliance

- **CMG measures: filling pressure, sensation, involuntary detrusor contractions (IDCs), compliance and capacity**.
- **Sensation is the only truly subjective parameter** — requires an alert patient and attentive clinician. ICS sensation categories (e.g. "early filling sensation", "urgency") are now considered **more clinically important** than previously; some treatments act specifically on sensation.
- **End of filling:** marked by "**permission to void**" on the trace or by stopping the pump; should be a **strong but not uncomfortable desire to void**. Consider the bladder diary and prior PVR when setting the target.

Detrusor overactivity. **DO** = urodynamic observation of **IDCs during filling cystometry, spontaneous or provoked** (Fig. 107.4); often with urgency or UUI.

COMPARISON: DETRUSOR OVERACTIVITY SUBTYPES (ICS 2018 MALE TERMINOLOGY)

Subtype	Definition
Idiopathic (primary) DO (IDO)	No identifiable cause
Neurogenic (secondary) DO (NDO)	DO with evidence of a relevant neurological disorder
Non-neurogenic (secondary) DO	An identifiable non-neurological cause exists

- "**Detrusor hyperreflexia**" and "**detrusor instability**" are obsolete.
- **All forms look identical on the tracing** — the label comes from **history (neurological status)**, not the CMG.
- **Interpretation in context:**
 - ideally **DO reproduces urgency or UUI** — but it can be **significant without symptoms, particularly NDO**;
 - **test-induced DO** occurs in a minority of **healthy asymptomatic volunteers**, and far more often during **ambulatory UDS**;
 - **absence of DO does not exclude it** — **up to 50% of women with UUI show no DO**;
 - **ability to suppress DO may itself be significant:** in mixed urinary incontinence (MUI) with a **normal CMG**, SUI surgery cured **both** stress and urgency incontinence far more often than antimuscarinic treatment cured UUI.
- **Confirm the contraction is involuntary** — a confused patient may simply void on first desire. Record **volume and pressure** of contractions; **repeat at a slower fill rate** if symptoms are uncharacteristic; note **DO incontinence (DOI)**.
- **Characterise DO:** single vs multiple; **phasic, sporadic or terminal** (near capacity); **suppressible vs non-abortable**; leakage or **precipitant micturition**.
 - **Prognostic use:** in obstruction with OAB, symptoms are **more likely to resolve after transurethral resection of the prostate (TURP)** when **DO is a single terminal IDC** than when it is continuous or sporadic.

- **Position:** DO is **more frequent in the vertical position** or on moving from supine to vertical → **test in the position that provokes the symptoms.**
- **NDO and upper-tract risk:** **higher maximum detrusor pressure (Pdetmax)** is associated with **worse renal function**, and **longer IDC duration** with upper-tract deterioration.
- Recognise **test-retest variation** — "usual LUT behaviour" is not always reproduced.

Compliance. Viscoelasticity (smooth muscle, collagen, elastin) makes the normal bladder **highly compliant** — little pressure rise with filling.

- **Compliance** = $\Delta V / \Delta P_{det}$, in mL/cm H₂O. ICS standard points: **Pdet at the start of filling (usually zero)** and **Pdet at cystometric capacity, or before any contraction causing significant leakage** — both excluding detrusor contractions.
- **No agreed normal value:** healthy means span a wide range, with great individual variation, because **compliance depends on capacity.**
- **Impaired compliance:** definitions of 10–20 mL/cm H₂O have been used; < 20 mL/cm H₂O has been proposed as indicating a poorly accommodating bladder — but a **small-capacity bladder can produce a low number without danger** (Fig. 107.6).
- **Absolute pressure is more useful than the compliance number:** **storage pressure > 40 cm H₂O is associated with upper-tract damage** (McGuire, 1981; Fig. 107.5). **Prolonged high-pressure storage causes upper-tract deterioration.**
- **Conventional cystometry can overstate pressure:** some neuropathic bladders with severely impaired compliance on CMG are **normal on ambulatory monitoring with natural filling.**
- **Causes of impaired compliance:**
 - **neurological** — SCI, spina bifida: from **increased outlet resistance (e.g. DESD)** or **decentralisation in lower motor neuron (LMN) lesions;**
 - **long-term BOO** (benign prostatic obstruction (BPO));
 - **structural** — **radiation cystitis, tuberculosis.**
- **Impaired compliance with prolonged high storage pressure is a urodynamic risk factor** and usually needs treatment to prevent renal damage (Box 107.1).
- **Pitfalls in measuring compliance:**

Pitfall	Mechanism	Resolution
Rapid filling	Non-physiological fill rate → accommodation problem , not true loss of compliance	Stop or slow filling and observe
Sustained low-amplitude IDC	Mimics compliance loss	Stop filling: pressure returns to baseline → compliance not impaired
Pop-off: vesicoureteric reflux (VUR)	Pressure transferred to the kidney; refluxed volume can exceed bladder volume → compliance looks falsely good	VUDS
Pop-off: bladder diverticulum	Part of the bladder — may genuinely protect upper tracts	VUDS
Pop-off: incompetent outlet	Leakage vents pressure; revealed only when outlet resistance rises	Occlude outlet during CMG — or unmasked only after artificial urinary sphincter or sling

Leak Point Pressure

- **Leak point pressure (LPP):** pressure (spontaneous or provoked) that expels fluid at the moment it is **visible outside the urethra** (also extraurethral loss or stoma). Record **provocation and recording site.**

- Practically there are **two distinct, independent types** — **ALPP and detrusor leak point pressure (DLPP)** — measuring completely different things; **both occur only in incontinent patients**.

Abdominal leak point pressure. Sphincteric "strength" — ability to resist abdominal pressure. Definition: **intravesical pressure at which leakage occurs from raised abdominal pressure in the absence of a detrusor contraction**.

- **Only demonstrable in SUI; no normal value** — continent patients do not leak at any physiological abdominal pressure. **Lower ALPP = weaker sphincter**.
- **Measure total pressure, not the change** — standing, include the **baseline (typically 20–40 cm H₂O)**. Read classically from **Pves**, provided there is no IDC (Fig. 107.7).
- **Volume**: originally **150 mL**; ALPP may **fall at higher volumes**, so record the volume. Practical sequence: **test at 150 mL → repeat every 50 mL until SUI appears → no SUI at capacity: remove the urethral catheter and read ALPP from the rectal line** (provided no Pdet rise from DO or poor compliance).
- **ALPP and intrinsic sphincter deficiency (ISD)** — McGuire's 1993 series of women with SUI:

ALPP THRESHOLDS FOR ISD (MCGUIRE, 1993)

ALPP (cm H ₂ O)	Clinical picture	Inference
< 60	High-grade incontinence, often continuous; mostly fixed urethra	ISD
60–90	Mostly marked hypermobility , moderate–high-grade leakage	Equivocal — component of ISD
≥ 90	Lesser grades; minimal to gross hypermobility	Little or no ISD

- **Use with caution:**
 - no technique separates **ISD from hypermobility** when both are present (hypermobility = **Q-tip deflection > 30°** from baseline);
 - **no hypermobility → SUI must be ISD, whatever the ALPP;**
 - **hypermobility is equally common at low and high ALPP** — ISD and hypermobility **coexist and are not discrete classes;**
 - **ALPP in isolation has limited value in predicting success** of common female SUI procedures.
- **ALPP ≠ VLPP**. ALPP may be provoked by **Valsalva (VLPP)** or **cough (cough leak point pressure, CLPP)** — in the same person **VLPP is significantly lower than CLPP**, so state the method.
- **Catheter effects: larger urethral catheter → lower ALPP** in women; ALPP can be read **without a urethral catheter** from the rectal (or vaginal, stomal) line. A proportion of women with SUI — and a larger proportion of **men with post-prostatectomy SUI** — **leak only with the urethral catheter removed**.
- **Newer inclusive term** — **urodynamic stress test**: any physical effort raising abdominal pressure during cystometry to examine SUI; report **provocation, catheter size/method, leak detection and volume**.

Detrusor leak point pressure. A measure of **Pdet in a poorly compliant bladder**: the **lowest Pdet at which leakage occurs in the absence of a detrusor contraction or raised abdominal pressure** (Fig. 107.5).

- **The higher the urethral resistance, the higher the DLPP**. Low outlet resistance → leakage at a low, "safe" pressure; **high outlet resistance → pressure keeps rising → less leakage but transmission to the upper tracts** (Fig. 107.8).
- **Assess DLPP in: upper motor neuron (UMN) lesions** with high storage pressure (DO + DESD); **LMN decentralisation; non-neurogenic low compliance** (multiple bladder operations, radiation, tuberculous cystitis).
- **McGuire (1981)**: in myelodysplastic children, **DLPP > 40 cm H₂O → hydronephrosis or VUR in 85%**.

- **But 40 cm H₂O is not a validated "safe" cut-off:** evidence was **low-level retrospective**; in myelodysplastic children **20 cm H₂O is more sensitive** for upper-tract damage; the **ICS 2017 teaching module** states only that **lower DLPP does better than higher**; a systematic review in neurological patients names simply **"high DLPP"** — with no cut-off — as a risk factor, alongside other factors such as compliance.
- **Treatment target:** as low a pressure as reasonably achievable.
- **Why an elevated DLPP matters:** bladder pressure is already too high **before the urethral "pop-off"** occurs. Treatment usually **lowers bladder pressure so DLPP is never reached**, or **lowers outlet resistance** (e.g. **sphincterotomy for DESD**) — then **repeat UDS to confirm**.

ALPP	DLPP
Measures the sphincter's response to abdominal pressure	Measures the injured bladder's response to outlet resistance
Low = bad (weak sphincter)	High = bad (danger to upper tracts)
Pressure source: abdomen , no detrusor contraction	Pressure source: detrusor (poor compliance) , no contraction or strain
Relevant in SUI	Relevant in low compliance / neurogenic bladder

- **DO leak point pressure** (ICS 2017 adult NLUTD terminology): **lowest Pdet rise with DO at which leakage first occurs**, without voluntary contraction or abdominal strain; clinical significance unexplored.

Stress-Induced Detrusor Overactivity

- **Stress-induced DO (SIDO):** a rise in abdominal pressure **triggers an IDC** — the symptom looks like SUI but the cause is **detrusor, not sphincteric** (Fig. 107.9).
- Note whether there is **also urodynamic SUI and/or DO independent of SIDO**.
- ICS equivalent: **cough-associated DO** — DO (with or without leakage) **immediately after the cough peak**.

Occult Stress Incontinence

- **SUI on prolapse reduction** (occult or latent SUI): SUI shown in a **clinically continent woman with POP only when the prolapse is reduced** — by **pessary, packing, Scopette, forceps or manually**. Shown on UDS = **urodynamic occult SUI**.

Urethral Pressure Profilometry

- **Brown and Wickham (1969)** method: small catheter with **lateral side holes, continuously perfused, slowly withdrawn**; bladder and urethral pressures recorded simultaneously. The transducer registers the pressure needed to **lift the urethral wall off the side holes**. **Accurate only where the urethra is distensible and seals** round the catheter.
- **Clinical relevance is controversial**; many urologists do not perform it routinely. Parameters (Fig. 107.10):
 - **urethral closure pressure (UCP) profile** = urethral pressure – intravesical pressure;
 - **maximum urethral pressure (MUP)** = highest pressure along the profile;
 - **maximum urethral closure pressure (MUCP)** = maximum difference between urethral and intravesical pressure;
 - **functional profile length** = length over which urethral pressure exceeds intravesical pressure (women).
- **Continent women:** functional length **~3 cm, MUCP 40–60 cm H₂O** — wide normal variation.
- **MUCP and ISD:** McGuire (1981) found **MUCP ≤ 20 cm H₂O predicted failure of SUI surgery** — the **fixed, open urethra ("type III" SUI)**, renamed **ISD in 1992**. **MUCP ≤ 20** shares the problems of ALPP-based ISD definitions.

- **ALPP vs MUCP: UPP does not diagnose SUI, and SUI is not required to measure it** — contrary to ALPP.
- **MUCP is lower in incontinent women but overlaps** with continent women and **does not grade severity**.
- **ICS standardisation (2002), unchanged since:** urethral pressure measurement has **unclear clinical utility** — it does not discriminate urethral incompetence from other disorders, does not grade severity, does not reliably predict surgical success, and does not normalise after surgery; **no newer technique has reached clinically relevant sensitivity, specificity or reliability**.

Figure Legends — Filling and Storage

- **Fig. 107.6 — The compliance paradox:** (left) steady rise to 50 cm H₂O at 400 mL = 8 mL/cm H₂O, with prolonged storage above 40 cm H₂O — **dangerous**; (right) **small-capacity bladder**, IDC at 40 mL with pre-contraction Pdet 5 cm H₂O — the **same calculated compliance (8 mL/cm H₂O)** but **not dangerous**, merely highly symptomatic.
- **Fig. 107.8 — Outlet resistance drives compliance:** two children with spina bifida, same neurology and symptoms. (A) **Poor outlet resistance** → low-pressure system, stress leakage, **low DLPP** — **upper tracts protected**. (B) **Strong outlet** → high-pressure system, **high DLPP**, **no stress leakage** — **upper tracts at risk**.

Concept — *Filling-phase interpretation is the act of **subtracting the abdomen and then asking what the detrusor did:** contract (DO) or fail to accommodate (compliance). The number that predicts harm is **absolute pressure over time**, not the label of the contraction or the computed compliance ratio.*

Voiding and Emptying Phase

Normal Voiding and Emptying

- The voiding phase assesses **detrusor contractility and outlet resistance**: the bladder must contract enough to overcome the outlet.
- **Abnormal emptying** = **outlet overactivity** (too much resistance), **detrusor underactivity** (weak force, short duration, slow velocity), or **both**. **PFS is the most accurate way to assess both** (Fig. 107.11).
- **Normal voiding function:** flow and pressure rise **within normal limits**, voiding starts promptly after permission and **ends with an empty bladder**. Male terminology: **normal detrusor contractile function** — an adequate continuous contraction emptying the bladder within a normal time.
- **Women may void normally with little Pdet rise** → the female definition differs.
- **Commonest reason normal voiding is not seen: increased outlet resistance** → **bladder outflow obstruction (BOO)**, a pressure-flow-based cut-off of clinically relevant resistance.
- **Micturition reflex, in sequence** (Fig. 107.12): **striated sphincter relaxes** → **detrusor contracts** → **bladder neck and urethra open** → **flow begins**. Coordinated by the **pontine and sacral micturition centres** with suprapontine voluntary control.

COMPARISON: ICS DETRUSOR FUNCTION DURING VOIDING

Term	Definition
Normal detrusor function	Voluntarily initiated continuous contraction → complete emptying within a normal time, without obstruction
Detrusor underactivity	Contraction of reduced strength and/or duration → prolonged emptying and/or incomplete emptying in a normal time
Acontractile detrusor	No demonstrable contraction during UDS
Neurogenic acontractile detrusor	Replaces " areflexia " when the cause is neurological

Term	Definition
Situational inability to void as usual	Attempted void judged unrepresentative — not "acontractile"

- **Detrusor underactivity is "nebulously defined"**: what constitutes reduced strength, duration or prolonged emptying is subjective, and **voiding pressure depends on outlet resistance** — correlate with the clinical picture.
- **Flow and PVR** show emptying is incomplete but **not why** (obstruction vs impaired contractility).
- **Qmax depends on VV: linear above 150 mL, hyperbolic below** → a **minimum VV of 150 mL** is widely recommended — a major limitation, as many patients never void that much. Alternatives: **corrected Qmax** ($Q_{\max} / \sqrt{VV}$) and **nomograms** correcting for VV — **Siroky (men)** and **Liverpool (men and women)**.

Bethanechol supersensitivity test — **Lapides (1962)**, applying **Cannon's law of denervation** (denervated structures become supersensitive to chemical stimulation) to distinguish **neurogenic from myogenic** underactivity.

- **Sequence**: fill at **1 mL/s to 100 mL**, record pressure (up to three times, averaged) → **bethanechol 2.5 mg subcutaneously** (later 0.035 mg/kg) → repeat at **10, 20 and 30 minutes**.
- **Negative (normal or myogenic)**: rise < **15 cm H₂O** above control at 100 mL at 30 minutes. **Positive (sensory or motor paralytic bladder)**: rise ≥ **15 cm H₂O**.
- **Limited role: unreliable** (poor sensitivity and specificity); treatment is usually the **same whatever the result (CIC)**; **bethanechol is not a consistently effective treatment** for underactivity, and a positive test **does not predict therapeutic response**.

Voiding Pressure–Flow Studies

- Begins at **cystometric capacity**, same bladder and rectal (or vaginal, stomal) lines, simultaneous uroflowmetry; ideally a **voluntary void**.
- **Pelvic floor contraction to stop leakage during an IDC** should be **annotated**; the **public, instrumented setting** inhibits some patients — account for this.
- **Pdet during voiding is a function of outlet resistance**: a normal detrusor meets obstruction with **higher pressure and lower flow** and still empties; over time it may **decompensate** → incomplete emptying or retention.

PRESSURE–FLOW PATTERNS

Pdet	Flow	Interpretation
Low (or normal)	High (or normal)	Normal, unobstructed
High	Low (or normal)	Obstruction
Low	Low	Detrusor underactivity — obstruction may coexist but be undiagnosable if the detrusor is too decompensated

- **BOO = high-pressure, low-flow voiding**; long-standing BOO may end in **detrusor underactivity**.
- **From rigid tubes to dynamic relations**: early resistance equations (from **1962**) used rigid-tube hydrodynamics, ignoring the **active, distensible urethra** and **bladder volume**, and were abandoned.

Bladder output relation (BOR) — Castro and Griffiths, 1972	Urethral resistance relation (URR) — Griffiths, 1975
Pressure–flow relation of the bladder, independent of the urethra , at a given volume	Pressure–flow relation of the urethra, independent of the bladder
Weakening contraction shifts the curve down	No flow until bladder pressure equals intrinsic urethral pressure , then flow rises rapidly; urethral relaxation moves the URR left , tightening right

- **Actual Pdet and flow = intersection of BOR and URR** (Fig. 107.14): changing one relation moves the point **along the other curve**.

Stop test — for suspected underactivity, estimates **isovolumetric detrusor pressure (Piso)**:

Voluntary stop test	Mechanical stop test	Continuous occlusion test
Patient interrupts flow midstream with the external sphincter	Balloon pulled against the bladder neck midstream, or urethra clamped/squeezed	Outflow occluded before contraction ; patient voids against it
Gives the lowest Piso	—	—

- The three **correlate well** in men and women.

Bladder Outlet Obstruction and Detrusor Underactivity in Males

- Premise: **surgical outcomes for BPH-related LUTS are better when obstruction is documented**; BPH/BPO's high prevalence made it the **model for defining obstruction**.

COMPARISON: MALE PRESSURE–FLOW NOMOGRAMS

Nomogram	Basis	Output
Abrams–Griffiths (1979)	PdetQmax vs Qmax	Obstructed / equivocal / unobstructed
URA — group-specific urethral resistance factor (Griffiths, 1989)	Urethral resistance	Obstructed / equivocal / unobstructed
Linear passive urethral resistance relation (LinPURR; Schäfer) (1990)	Linearised URR	Grades of obstruction (incl. "slightly obstructed") and contractility
ICS provisional nomogram (Griffiths, 1997)	BOOI	Recommended for men with LUTS suggestive of BPH

- All derived from **men undergoing surgery (mainly TURP)**: after surgery, **pressure at maximum flow (PdetQmax)** fell in the obstructed, fell **unpredictably** in the equivocal, and was **unchanged** in the unobstructed. All three classify patients similarly.
- **Bladder outlet obstruction index (BOOI)** (the "AG number") — the slope of the **Abrams–Griffiths obstructed/equivocal line**, identical to Schäfer's obstructed/slightly obstructed line:
 - $BOOI = PdetQmax - 2Qmax$;
 - **> 40 obstructed • 20–40 equivocal • < 20 unobstructed** (Fig. 107.15).
- **Reproducibility is good**: random variation of roughly **9–14 cm H₂O** in pressure and **0.4–2 mL/s** in Qmax; repeated studies in one session show a small **systematic fall** in Pdet and Qmax. Only a minority change ICS class, **almost always by one class only**.
- **Bladder contractility index (BCI)** — from Schäfer's contractility groups (strong, normal, weak, very weak):
 - $BCI = PdetQmax + 5Qmax$;

- **> 150 strong • 100–150 normal • < 100 weak** (Fig. 107.16);
- **the most widely used contractility measure** — easy to calculate, linked to BOOI and the ICS nomogram.

BOOI	BCI
$P_{det}Q_{max} - 2Q_{max}$	$P_{det}Q_{max} + 5Q_{max}$
> 40 obstructed; < 20 unobstructed	> 150 strong; < 100 weak
Measures outlet	Measures detrusor

- **Obstruction can be invisible in an underactive detrusor:** unequivocal obstruction needs **BOOI ≥ 40** , i.e. **$P_{det} \geq 40$ cm H₂O even at zero flow**; at Q_{max} 5 mL/s, **$P_{det}Q_{max} \geq 50$** . The ICS nomogram therefore misses men whose weak detrusor is itself the result of long-term obstruction → clinical judgement.

Watts factor (WF) — **detrusor power per unit bladder surface area**, corrected for the power needed for isometric contraction or unloaded shortening:

- **$WF = [(P_{det} + a)(v_{det} + b) - ab] / 2\pi$** , where v_{det} = **detrusor shortening velocity**, $a \approx 25$ cm H₂O, $b \approx 6$ mm/s.
- WF varies through the void; representative points are **maximum power (W_{max})** and **power at maximum flow (WQ_{max})**.

Advantages	Disadvantages
Minimally dependent on bladder volume	No measure of sustainability of contraction
Unaffected by BOO	Complex calculation — limited clinical use
	Normal threshold not validated (W _{max} 7.0 W/m ² suggested)

- **BOO also disturbs storage** (ultrastructural change): **about two-thirds of men with symptomatic BPO have DO**, which resolves in half to two-thirds after relief of obstruction; reduced compliance also improves after TURP.

Micturitional urethral pressure profile (MUPP) — voiding profilometry (Yalla): **both diagnoses and localises obstruction**.

- **Technique:** triple-lumen catheter under fluoroscopy → **slowly withdrawn during voiding** from bladder neck to anterior urethra.
- **Normal:** bladder **isobaric with the prostatic urethra**, then a **20–30 cm H₂O drop across the membranous urethra** (the **narrowest segment during voiding**), then gradual decay.
- **BPH obstruction:** a **pressure drop > 5 cm H₂O within the prostatic urethra** marks obstruction at that point.
- **As effective as standard PFS** in diagnosing BOO; treatment based on MUPP has given successful outcomes.
- **Piso in men:** when PFS cannot separate obstruction from underactivity (inefficient emptying, neither diagnosis made), **Piso < 50 cm H₂O is uncommon and considered diagnostic of detrusor underactivity**.
- **Contraction-strength methods compared (LinPURR, BCI, W_{max})** — all correlate well; **LinPURR is simple and intuitive and assesses contraction and BOO together**; **W_{max}** suited to studies aiming to improve contractility.
- **No universal criteria for detrusor underactivity:** UDS is "required" for the diagnosis, yet ICS gives **no standard for normal contraction duration or normal micturition time**.

Bladder Outlet Obstruction in Females

- **More of a diagnostic dilemma:**
 - **no highly prevalent cause** equivalent to BPH to build nomograms on;
 - **male nomograms do not apply** — voiding dynamics differ;
 - many women **void by pelvic floor relaxation alone**, some by **straining**;
 - **small Pdet rises or flow falls**, insignificant in men, may signify BOO.
- **High index of suspicion:** LUTS, incomplete emptying, **persistent UTIs**, previous **anti-incontinence surgery**, prolapse.

COMPARISON: FEMALE BOO CUT-POINT CRITERIA

Study	Controls	Qmax	PdetQmax
Chassagne (1998)	Women with SUI	≤ 15 mL/s	≥ 20 cm H ₂ O
Lemack and Zimmern (2000)	Expanded population	≤ 11 mL/s	≥ 21 cm H ₂ O
Defreitas (2004)	Normal asymptomatic women	≤ 12 mL/s	≥ 25 cm H ₂ O

- **Limitation:** obstruction was **predefined clinically** and **anatomic only** — **functional obstruction** (primary bladder neck obstruction (PBNO), dysfunctional voiding) was excluded. **Anatomic and functional obstruction share voiding pressures and overlap widely in flow** → future definitions need not exclude either.

COMPARISON: METHODS OF DIAGNOSING FEMALE BOO

Method	Definition	Strength	Weakness
Cut-points (above)	Qmax and PdetQmax thresholds	Simple	Anatomic cases only; large overlap
VUDS criteria (Nitti, 1999)	Radiographic obstruction between bladder neck and distal urethra with a sustained detrusor contraction of any magnitude	Localises; diagnoses obstruction despite impaired contractility	Needs fluoroscopy
Blaivas–Groutz nomogram (2000)	Free (non-invasive) Qmax vs Pdetmax; four zones: none, mild, moderate, severe	Uses the higher uncatheterised flow	Built from two separate voids; overdiagnoses obstruction
Solomon–Greenwell nomogram (2017)	VUDS as referent; axis PdetQmax = 2.2 × Qmax; probability of obstruction	Includes functional and anatomic obstruction	Newer, less validated

- **VUDS and cut-point criteria show good concordance; Blaivas–Groutz overdiagnoses.**
- **Absolute pressure–flow values overlap** between obstructed and unobstructed women → **a second parameter (radiographic or clinical) is needed.**
- **Solomon–Greenwell:** $PdetQmax - 2.2 \times Qmax < 0$ → low chance (< 10%); > 5 → likely (> 50%); > 18 → almost certain (> 90%).
- **The ICS nomogram and BOOI cannot be used in women** — they **grossly underestimate female BOO**, because women normally void at **much lower pressures**. Concept the same (**higher pressure, lower flow**); values different and less well defined.
- **Piso in women:** not validated for characterisation; **significantly lower in older women than in men**; in older women with UII, **voluntary stop test gives the lowest value**, mechanical and continuous occlusion higher.

- **Female detrusor underactivity:** no universal criteria; women with underactivity (no voiding contraction > 10 cm H₂O, voiding by straining or unable to void) have **reduced bladder sensation — contractility and sensation are inversely associated.**

Concept — Every male index is built on the **BPH model**, which is why it fails in women: without a common obstructing disease and with physiologically low-pressure voiding, female obstruction can only be defined by **adding a second, non-pressure criterion** — ideally an image of where the outlet narrows.

Sphincter Coordination

The External Sphincter

- **Normal voluntary voiding:** urethral pressure falls and EMG quietens before Pdet rises; sphincter stays relaxed until voiding ends, then activity returns (Fig. 107.13). **Failure to relax or stay relaxed is abnormal.**
- **Guarding reflex:** EMG rises with an involuntary contraction to inhibit the IDC — **normal**, not dyssynergia.

COMPARISON: DYSSYNERGIA VS DYSFUNCTIONAL VOIDING

DESD	Dysfunctional voiding
Involuntary external sphincter contraction during DO and voiding	Learned failure to relax or involuntary contraction of the external sphincter during voiding
Requires a neurological lesion above the sacral micturition centre (suprasacral cord); the higher the lesion, the more likely	No neurological lesion
High pressures → impaired compliance over time → urodynamic risk factor for upper-tract deterioration (Box 107.1)	Described in children; also adult men and women — a major cause of LUTS; storage symptoms common
—	Confirm the intermittent/reduced flow on free uroflowmetry to exclude a test artefact; fluoroscopy improves accuracy

- **Adult male diagnostic criteria for dysfunctional voiding:** no neurological history or signs → **transient, intermittent external sphincter closure during voiding on EMG and fluoroscopy → raised sphincter EMG without abdominal pressure rise → discontinuous free flow.**
- **"Non-relaxing" sphincter** → reduced flow; **delayed relaxation of the urethral sphincter** → delayed flow onset, in adult neurological dysfunction.

The Internal Sphincter

- **DISD:** a suprasacral lesion **above the sympathetic ganglia (T10–L1)** may cause **DISD together with DESD.**
- **Bladder neck dyssynergia / PBNO:** a recognised cause of LUTS in **non-neuropathic men, women and children;** aetiology unknown.
- **Internal sphincter dysfunction requires VUDS for diagnosis.**

KEY POINTS

- **Pdet near zero throughout filling;** subtraction separates detrusor from abdominal rises.
- **Confirm contractions are involuntary;** usual behaviour is not always reproduced.
- **Prolonged high-pressure storage damages the upper tracts.**
- **Low ALPP = weak sphincter; high DLPP = dangerous outlet resistance.**
- **BOO = high-pressure, low-flow voiding;** underactivity = reduced strength and/or duration.

- **EMG rise with an IDC is a normal guarding reflex; true DESD requires a suprasacral lesion.**

Concept — *The same sphincter behaviour — failure to relax during voiding — is named by its **cause, not its tracing**: with a suprasacral lesion it is DESD and a threat to the kidneys; without one it is dysfunctional voiding, a learned behaviour. The bladder neck is invisible to EMG, so its dysfunction can only be seen on fluoroscopy.*

VIDEOURODYNAMICS

- **VUDS = simultaneous UDS and fluoroscopy** — the **most precise evaluation** of voiding function, especially where **anatomy and function interact**.
- **Uses:**
 - **localisation of obstruction;**
 - **incontinence not seen on examination;**
 - **VUR during storage and/or voiding;**
 - **NLUTD;**
 - **the only way to evaluate bladder neck function;**
 - **confirms sphincter dysfunction** diagnosed by EMG, or diagnoses it when EMG is equivocal;
 - determines whether a **known anomaly** (bladder or urethral diverticulum, VUR) contributes.
- **Equipment:** usually a **C-arm** (most flexible), or a fixed tilting table (90–180°).
- **Position for the question:** **SUI** — **standing** (both sexes); **voiding** — **usual position** (women sitting, men standing).
- **Limit screening time** to high-yield moments: **provocative manoeuvres, pressure rises (compliance loss, IDCs) and voiding.**

Functional obstruction in women. VUDS criteria (Nitti, 1999): **radiographic obstruction between bladder neck and meatus during voluntary voiding defines and localises obstruction. PBNO can only be diagnosed on VUDS.** The two commonest functional causes are **PBNO (Fig. 107.18)** and **dysfunctional voiding (Fig. 107.19).**

- **VUDS is more accurate than surface EMG** for diagnosing dysfunctional voiding and separating it from PBNO — surface EMG alone **misdiagnoses a substantial minority** of dysfunctional voiders, and some PBNO patients show raised EMG during voiding.

Young men — **PBNO**. PFS alone can show obstruction, but surgeons want it **localised** before operating on a young man; PBNO may lack the classic pattern. Three types:

Type	Pattern
I	Classic high-pressure, low-flow
II	Normal pressure, low flow, with narrowing at the bladder neck
III	Delayed opening of the bladder neck

- **Fluoroscopy is critical, especially for types II and III;** it also localises BPO, bladder neck contracture and urethral stricture once urodynamic obstruction is confirmed.
- **AUA/SUFU UDS guideline:** VUDS supported in **young men and women without an obvious anatomical cause** — separates **PBNO from dysfunctional voiding**; the **only tool that documents pressure–flow and localises functional bladder neck obstruction. No outcome comparison** with non-VUDS-directed treatment exists.

Neurogenic and high-pressure bladders.

- **VUR:** record **volume and pressure at onset**, side and grade. **Reflux is a pop-off:** without fluoroscopy, **measured compliance overestimates true compliance** — a potentially unsafe error (Fig. 107.21).
- **DLPP can be observed** in patients who cannot be positioned to see leakage (e.g. some tetraplegics).
- **DISD — often with DESD — can only be diagnosed by VUDS**, and changes treatment (Fig. 107.22).

COMPARISON: GUIDELINE POSITIONS ON VUDS IN NLUTD

European Association of Urology (EAU)	AUA/SUFU
VUDS is the optimum test for neuro-urological disorders	VUDS adds worthwhile information beyond CMG or PFS alone
Unavailable → filling CMG plus PFS	Radiation is additive → lowest dose that answers the question

- **Balance:** diagnostic gain in selected populations vs **radiation exposure, cost and limited availability.**
- **VUDS is the diagnostic tool of choice for bladder neck dysfunction in both sexes.**
- **Refer to a VUDS centre** when diagnosis is otherwise incomplete in high-risk patients: **known or suspected NLUTD, unexplained retention in women, previous radical pelvic surgery, urinary diversion, pre- or post-renal transplant, prior pelvic radiation.**

Figure Legends — Voiding Phase and Videourodynamics

- **Fig. 107.18 — PBNO** (35-year-old woman, retention): **bladder neck fails to open despite a contraction > 60 cm H₂O.** (Nitti, 2005.)
- **Fig. 107.19 — Dysfunctional voiding** (23-year-old woman): IDC before voiding, **raised EMG during voiding**, "spinning-top" urethra with obstruction at the external sphincter; high-pressure, low-flow voiding.
- **Fig. 107.21 — Reflux pop-off:** 75-year-old man with raised PVR and left hydronephrosis; **end-filling pressure only 20 cm H₂O ("safe")**, but VUDS shows **reflux from low pressure** — true functional compliance is **far worse** than pressure suggests.

KEY POINT

- **VUDS is the gold standard for invasive UDS in NLUTD.**

Concept — *Fluoroscopy adds the one thing pressure lines cannot supply: **where**. It localises obstruction, reveals the bladder neck, and exposes pop-off reflux that makes a dangerous bladder look safe — which is why it is indispensable in neurogenic and functional-obstruction cases and optional elsewhere.*

CLINICAL APPLICATIONS OF URODYNAMICS: EVIDENCE-BASED REVIEW AND GUIDELINES PERTAINING TO URODYNAMICS

- For many situations evidence is insufficient and use rests on **clinical impression**; for some, evidence is strong enough to guide use. Three areas are reviewed: **women with SUI, men and women with LUTS, and NLUTD.**

Guidelines

- **AUA/SUFU UDS guideline (2012): 19 statements** across **four disease states** — SUI/POP, OAB/UUI and MUI, neurogenic bladder, LUTS. **Not reconfirmed since**; guidelines encourage best practice but **do not establish the standard of care.**

- **Shift to disease-specific guidelines: AUA/SUFU surgical SUI guideline (2023) — two UDS statements;** AUA/SUFU OAB and bladder pain syndrome/interstitial cystitis guidelines; EAU urinary incontinence guideline (originally six UDS statements for neurologically intact adults; now separate **male and female** guidelines).
- **Many guidelines are redundant**, built on the same data.
- **Oxford grading** (used by AUA UDS guideline):

Grade	Evidence
A	Consistent level 1 — effectively mandatory, placed in care pathways
B	Consistent level 2 or 3 , or majority evidence from RCTs
C	Level 4 , or majority evidence from level 2/3, or expert opinion

- Guidelines **do not guarantee better outcomes**, do not replace clinical expertise, and are **not a legal standard**.

Evaluation of Females With Stress Urinary Incontinence

- For **pure SUI** — no urgency, normal emptying, SUI demonstrated on examination — **UDS adds little**.
- **SISTER trial** (Burch vs pubovaginal sling, pure or stress-predominant MUI, no obvious emptying problem): UDS (ALPP, DO) **did not help predict efficacy or postoperative voiding dysfunction**.
- **NICE (2006)**: UDS before SUI surgery **only if** — suspected **DO**, **previous SUI** or **anterior-compartment prolapse surgery**, or **symptoms of voiding dysfunction**.

COMPARISON: RANDOMISED EVIDENCE ON PRE-OPERATIVE UDS IN UNCOMPLICATED SUI

	ValUE trial (Value of Urodynamic Evaluation)	Dutch multicentre cohort with embedded non-inferiority RCT
Population	Uncomplicated pure or stress-predominant MUI; PVR < 150 mL; negative urinalysis/culture; urethral mobility and positive stress test ; excluded prior incontinence surgery, pelvic radiation, recent pelvic surgery, significant anterior/apical POP	SUI or stress-predominant MUI; all had UDS ; those with UDS discordant from clinical assessment randomised
Comparator	Office evaluation + UDS vs office evaluation alone (630 women)	Immediate mid-urethral sling (MUS) vs UDS-tailored treatment
Primary endpoint	UDI reduction ≥ 70% and PGI-I "much / very much better" at 12 months	Improvement on UDI incontinence subscale
Result	Success 76.9% vs 77.2% (difference −0.3 points , 95% CI −7.5 to 6.9)	Nearly half had discordant UDS ; most tailored-arm patients still had surgery; subjective and objective cure equal
Verdict	Office evaluation alone non-inferior	Immediate MUS non-inferior to tailored therapy

- **ValUE detail**: UDS changed the sling route in a few women (transobturator ↔ retropubic), with unknown effect; UDS **increased surgeon confidence** in diagnosing ISD, OAB and voiding dysfunction — **but not SUI** — without improving outcome. (UDI = Urogenital Distress Inventory; PGI-I = Patient Global Impression of Improvement.)
- **Conclusion**: UDS is **not essential before surgery for stress-predominant incontinence when SUI is clinically demonstrated**. Its real role is in **mixed symptoms or emptying difficulty**, where symptoms predict urodynamic SUI, DOI and MUI **inconsistently**.

- **Urgency with urodynamic SUI and a normal CMG (no DO):** excellent cure of **both** stress and urgency symptoms after pubovaginal sling, Burch or tension-free vaginal tape. The "no difference" trials were largely **stress-predominant** — in **urgency-predominant** women DO may matter more.

COMPARISON: GUIDELINE STATEMENTS ON UDS IN FEMALE SUI

Body	Statement	Strength
AUA/SUFU UDS (2012)	Assess urethral function (ALPP or MUCP) when diagnosing urodynamic SUI	Recommendation; C
	Assess PVR before invasive SUI therapy	Expert Opinion
	May perform multichannel UDS in SUI before invasive, morbid or irreversible treatment	Option; C
	Repeat stress test with catheter removed if SUI not shown with it in place	Recommendation; C
	High-grade POP without SUI → stress test with prolapse reduction; multichannel UDS with reduction may assess occult SUI and detrusor dysfunction	Option; C
AUA/SUFU surgical SUI (2023)	May omit UDS in the index patient when SUI is clearly demonstrated	Conditional; B
	May perform UDS in non-index patients	Expert Opinion
EAU (neurologically intact incontinence)	Replicate symptoms; check recording quality; interpret in clinical context; remember intra-individual variability	Strong
	Do not routinely perform UDS for uncomplicated SUI	Strong
	Perform UDS if findings may change invasive treatment	Weak
	Do not routinely use UPP or LPP to grade severity	Strong

- **AUA non-index patients** (UDS at discretion): **prior anti-incontinence surgery; prior POP surgery; subjective-objective mismatch; significant voiding dysfunction; significant urgency, UUI or OAB; elevated PVR; unconfirmed SUI; NLUTD.**
- **Occult SUI and prolapse surgery:** two RCTs (with and without concomitant anti-incontinence surgery) show **more de novo SUI when occult SUI is present pre-operatively** → UDS with reduction can **risk-stratify** — but this is **debated**. The ICS notes **UDS informs counselling** but there is **no compelling evidence it determines surgical outcome**. **Office evaluation with reduction may perform as well;** UDS detects more SUI but **does not better predict post-operative SUI.**
- **"Guidelines of guidelines"** (AUA, EAU, NICE 2019, Canadian Urological Association): **invasive UDS not needed before conservative treatment;** may be used if it would **change management** (before invasive therapy); **agreement to use UDS for recurrent incontinence after failed invasive treatment;** considerable redundancy.

Concept — *In women with SUI, UDS is judged by one test: **does the result change the operation?** For demonstrable, uncomplicated SUI it does not, so it may be omitted; for mixed, recurrent, complex or non-index patients it may, so it is used.*

Evaluation of Males and Females With Lower Urinary Tract Symptoms

- **LUTS is multifactorial** — at least four conditions: **BOO, detrusor underactivity, DO and "sensory urgency"**. Storage and voiding symptoms coexist; **UUI** may come from DO with or without BOO.
- **LUTS are common in men ≥ 50; coexistence of BOO and DO rises with age and with BOO severity.**

- **PVR (AUA, Clinical Principle):** may be done as a **safety measure to exclude significant retention**, initially and at follow-up — cannot separate obstruction from underactivity, and **no agreed "elevated" threshold**.
- **Uroflow (AUA, Recommendation; C):** may be used in the initial and ongoing evaluation of **men** with voiding/emptying LUTS — likewise cannot distinguish obstruction from underactivity, but guides therapy and prompts testing.
- **How much diagnosis before treatment?:** **no UDS needed to start an α -blocker** in a man emptying normally; as treatment becomes **more invasive and morbid**, a precise diagnosis becomes more valuable.
- **After TURP, storage symptoms persist in about one-third.** When DO accompanies BOO, **single terminal IDC $\rightarrow$ storage symptoms more likely to resolve** than continuous or sporadic IDCs — valuable for **counselling** (a man bothered mainly by emptying may choose surgery; one bothered mainly by OAB may not).
- **Evidence for UDS before BPH surgery is mixed** — some support, some find it unnecessary or useful only in inconclusive cases, some find modest predictive value. **6th International Consultation on New Developments in Prostate Cancer and Diseases:** evidence is **level 3** but its quantity supports a **grade B recommendation**.
- **"No obstruction on PFS $\rightarrow$ less favourable surgical results."** No critical BOOI level for preventing decompensation has been defined, and **no evidence-based trigger for surgery to prevent bladder decompensation** exists — though BOO causes progressive, possibly **irreversible structural bladder change**, prompting some to favour early relief.
- **Significantly impaired compliance is the only "absolute urodynamic indication" for treating BOO.**

COMPARISON: GUIDELINE STATEMENTS ON PRESSURE–FLOW STUDIES IN MALE LUTS

Body	Statement	Strength
AUA/SUFU UDS (2012)	PFS when it is important to determine whether urodynamic obstruction is present, particularly before invasive, morbid or irreversible treatment	Standard; B
AUA BPH (2021)	PVR before intervention	Clinical Principle
	Consider uroflowmetry before intervention	Clinical Principle
	Consider PFS when diagnostic uncertainty exists	Expert Opinion
EAU non-neurogenic male LUTS (2023) — all Weak	PFS only for specific indications before invasive treatment or to study pathophysiology	Weak
	Previous unsuccessful (invasive) treatment	Weak
	Considering invasive treatment and cannot void > 150 mL	Weak
	Considering surgery, bothersome voiding LUTS, $Q_{\max} > 10$ mL/s	Weak
	Considering invasive therapy, voiding LUTS, PVR > 300 mL	Weak
	Considering invasive treatment, voiding LUTS, age > 80 years	Weak
	Considering invasive treatment, voiding LUTS, age < 50 years	Weak

- **What the EAU list is testing:** each trigger marks a man in whom **obstruction cannot be assumed** — flow good enough to cast doubt on BOO (**$Q_{\max} > 10$**), a volume too small to interpret, retention that may be detrusor failure (**PVR > 300**), extremes of age (**< 50** — other causes; **> 80** — underactivity), or failed prior treatment.
- **Trap:** the EAU **$Q_{\max}$ trigger is > 10 mL/s**, not < 10 — a low $Q_{\max}$ already suggests obstruction.

- Across guidelines PFS is generally **optional**; uroflow and PVR usually optional, occasionally "recommended" or mandatory.

Filling cystometry in LUTS. Impaired compliance can only be diagnosed by CMG — from BOO, radiation cystitis or inflammatory disease (tuberculosis) — and may justify treatment **independent of symptoms**; UDS is advised when suspected.

- **AUA/SUFU**: may perform multichannel filling cystometry to detect **altered compliance, DO or other abnormalities** in UUI before invasive, morbid or irreversible treatment (**Option; C**), and in LUTS generally (**Expert Opinion**).
- **AUA/SUFU OAB**: UDS **should not be used in the initial work-up of uncomplicated OAB**; latitude for complicated or refractory patients.
- **Women with poor emptying or voiding symptoms**: PFS separates obstruction from impaired contractility — most useful when the cause is **not obvious**. **Anatomic obstruction** (high-grade POP, post-incontinence surgery, urethral mass) is usually evident; **functional obstruction (dysfunctional voiding, PBNO)** is **not visible on examination or endoscopy** and needs voiding evaluation. **AUA/SUFU**: may perform PFS in women to determine obstruction (**Recommendation; C**); **VUDS** diagnoses and localises.

Concept — *Symptoms cannot tell obstruction from underactivity, so the value of PFS rises with the invasiveness and irreversibility of the planned treatment — and the one finding that mandates treatment regardless of symptoms is a bladder that stores at high pressure.*

Evaluation of Neurogenic Lower Urinary Tract Dysfunction

- **Presentations**: incontinence of **bladder origin** (DO, impaired compliance) or **sphincter origin**; poor emptying of **bladder origin** (neurogenic detrusor underactivity, neurogenic acontractile detrusor) or **sphincter origin** (dyssynergia); or **silent upper-tract decompensation** — hydroureteronephrosis and renal insufficiency **without bothersome symptoms**.
- **Goals**: **prevent upper-tract decompensation** and relieve symptoms; achieve physiological filling (and if possible voiding), control symptoms, and a management situation acceptable to the patient. **UDS has its most important role here**.
- **Evidence is mostly level 3 or lower** — **RCTs are considered dangerous and unethical** given the consequences of untreated neurogenic dysfunction.
- **Primary aim**: **convert to a low-pressure bladder during filling, even at the cost of incomplete emptying and catheterisation**. Therapy depends on whether the detrusor is **overactive or poorly compliant** — **only UDS answers this unequivocally**. UDS is also essential for **monitoring response and sequelae**.

Ice water test (IWT) — **Bors and Blinn (1957)**; no agreed protocol.

- **Mechanism**: cold instillation elicits a **C-fibre-mediated spinal reflex detrusor contraction**, normally **inhibited supraspinally**.
- **UMN lesion** → **inhibition lost** → **contraction (positive)**; **LMN lesion** → **no contraction (negative)**.
- Helps distinguish **reflexic from areflexic** bladders; **ICS: optional**, interpreted with other data, and **should not precede standard UDS**.

Who needs UDS.

UDS usually not needed first	UDS helpful	UDS essential (before treatment and in follow-up)
High storage pressure not suspected — UUI after stroke ; woman with MS and low PVR	Men with possible BPO plus Parkinson disease or MS	Potentially harmful storage — SCI , myelomeningocele

COMPARISON: GUIDELINE STATEMENTS ON UDS IN NLUTD

Test	AUA/SUFU UDS (2012)	EAU Neuro-Urology (2023)
PVR / non-invasive	PVR at initial evaluation of relevant neurological conditions (SCI, myelomeningocele) and in follow-up — Standard; B	Non-invasive testing mandatory before invasive UDS — Strong
CMG	Complex CMG at initial evaluation , with or without symptoms, and in follow-up; optional in other neurological disease with LUTS — Recommendation; C	UDS is the only objective assessment of LUT dysfunction (LE 2a); same-session repeat measurement — Strong
PFS	In relevant neurological disease with or without symptoms, or other neurological disease with raised PVR or symptoms — Recommendation; C	Filling cystometry continuing into PFS if VUDS unavailable — Strong
VUDS	When available, may perform in relevant disease at risk, or other disease with raised PVR or symptoms — Recommendation; C	VUDS is the optimum procedure (LE 4); use VUDS — Strong
EMG	With CMG ± PFS in the same groups — Recommendation; C	Specialised uroneurophysiological tests elective, specialised settings only (LE 4)
Technique	—	Physiological filling rate and body-warm saline — Strong

- The two bodies **agree**: non-invasive testing first, **VUDS preferred** in neurogenic patients; the EAU adds technical detail.

KEY POINTS

- **UDS is not essential before surgery for stress-predominant MUI** in women when SUI is clinically demonstrated.
- **AUA**: PFS to determine obstruction in men with LUTS is important before **invasive, morbid or irreversible** treatment.
- **Women with poor emptying or voiding symptoms** benefit from PFS to separate obstruction from impaired contractility, especially when the cause is not obvious.
- **UDS has its most important role in NLUTD.**

Concept — *In NLUTD the kidneys, not the symptoms, set the agenda: because damage can be silent and trials are unethical, UDS is the only arbiter of whether the bladder stores safely, and **low filling pressure outranks complete emptying**.*

FUTURE DIRECTIONS

- UDS remains a key **research tool**; future value lies in **shaping decisions, patient understanding and phenotyping**, and predicting long-term management.
- **Non-invasive tests for male BOO** (systematic review): **detrusor wall thickness, near-infrared spectroscopy (NIRS), penile cuff test, uroflowmetry with Qmax < 10 mL/s, intravesical prostatic protrusion > 10 mm** — promising but limited by **heterogeneous definitions and thresholds, small samples and weak methodology**; combinations may be needed.
- **Penile cuff test (UroCuff)**: **cuff pressure needed to interrupt flow** estimates bladder pressure; **no urethral catheter, naturally filled bladder**; plotted on a **modified ICS nomogram**. A meta-analysis found it **performs sufficiently** for diagnosing and managing BOO, but obstruction criteria vary.

- **Storage-phase surrogates for DO: ultrasound bladder/detrusor wall thickness and bladder weight; NIRS** (oxy-/deoxyhaemoglobin changes during IDCs in real time — **limited by motion artefact**); **biomarkers such as nerve growth factor (NGF)** (correlate with DO/OAB but **lack specificity**). None is reliable for routine practice or quantifies storage beyond DO → **filling cystometry remains the gold standard for storage disorders**.
- **Functional magnetic resonance imaging (fMRI)** with concurrent UDS: maps **brain network activation through the micturition cycle** (controls and OAB) — research only.
- **Implantable wireless bladder pressure monitors**: longer, out-of-office monitoring; mostly **animal data**, recent human work shows **promising reliability, safety and tolerability**, but a **single intravesical channel** needs validation.

Concept — *Each new technology tries to remove the catheter or the laboratory from urodynamics; none yet replaces the subtracted-pressure principle, which is why **filling cystometry and pressure-flow remain the reference standards** against which they are all judged.*